# Train, test, trust?

## AI · Level 3 · Lesson 3



# Today we will

- Explain overfitting, and why a model's score must come from examples it never trained on.
- Tell false positives from false negatives, and say which costs more in a school case.
- Question an accuracy claim: tested on whom, on which data, with which mistakes?

# Remember

- A perfect training score can hide overfitting: memorised examples, not a learned pattern.
- Only a score on a test set the model never saw counts, and only for people like those in it.
- Every model makes false positives and false negatives; ask which it makes, and which costs more.

# Memoriser and learner

1. Warm-up (5 min): Who has met a pupil who memorised past papers?
2. Memoriser and learner (12 min): Two pupils, ten old and five new questions.
3. Overfitting (6 min): Name what happened, and link it to the test set.
4. Two mistakes (9 min): Sort cards into false alarms and misses.
5. Wrap-up (8 min): The three questions, then the exit ticket.

**Prepare a card with ten rectangles and their areas (for example 3 by 4: 12), five new rectangles, and six mistake cards: three false alarms and three misses.**

# Let’s discuss

1. Where do pupils in our school memorise instead of learning?
2. In a dropout warning, is a false alarm or a miss worse? Why?
3. Which AI claim have you heard that deserves the three questions?

# Exit ticket

- Explain overfitting in one sentence, and write one false positive and one false negative from school life.